

GII Index Reduction Analysis Report

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1 Executive Summary

This report documents a method for determining how far the number of indicators in the Global Innovation Index (GII) 2025 can be reduced without significantly compromising the index’s discriminatory power. The method was previously applied to the EU JRC’s Sustainable and Inclusive Well-being Index (Benczur et al. 2024; Boskovic et al. 2024); this report replicates it on GII’s 78 Level-1 indicators observed for 139 countries/economies.

Two perspectives are evaluated: a **country perspective** (do countries end up the same distance apart using a subset as with the full indicator set?) and a **variable perspective** (does a subset preserve the PCA structure of the full set?). The two perspective rely on complementary metrics. The country perspective is based on a Spearman rank correlation between full- and subset-distance matrices; the variable perspective is based on r_m^2 - the percentage of the full PCA’s variance structure recovered by the subset.

Both perspectives require the analyst to choose a subset of indicators. Subsets of 21 and 42 indicators are considered. As the GII has 21 sub-pillars, 21 is the minimum needed to cover every concept the index aims to measure.

Random selection provides the baseline: the best subset drawn from 1000 random selections at a given size, ignoring the conceptual structure entirely. Against it, four structured allocation rules are evaluated. Flat allocation assigns the same number of indicators to every group, either 3 per pillar or 1 per sub-pillar, both yielding 21 indicators. Proportional allocation instead sets each group’s budget in proportion to its share of the index, on the premise that a group’s indicator count reflects its importance or the difficulty of measuring it; this is applied across pillars and across sub-pillars at 42 indicators, and across pillars again at 21 so that the two rules can be compared at a common budget. Proportional allocation across sub-pillars at

21 indicators is not evaluated separately, because that apportionment reduces to exactly one indicator per sub-pillar and is therefore identical to the corresponding flat rule.

The headline recommendation is that the “one indicator per sub-pillar” set of **21- indicators comes close to reproducing the PCA structure** ($r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full PCA’s variance structure) while also closely reproducing the distance between countries (Rand Index 0.887 against the full-data country clustering). A larger 42-indicator subset, built by allocating indicators proportionally to each pillar’s actual size rather than a flat count per pillar, reaches $r_m^2 = 92.6\%$ and a rank correlation of 0.955 with the full country-distance matrix. Both are documented below alongside the flat and proportional strategies they were chosen from.

Relative to random subsets of matched size, exhaustive within-group search yields a modest advantage when indicators are allocated at pillar level and none when they are allocated at sub-pillar level. This pattern holds at both the 21- and 42-indicator budgets, and under both metrics.

2 Introduction

When creating indicator dashboards and composite indices, indicator selection can be daunting. Stakeholders, statistics and conceptual relevance often diverge in recommendations or build up a large set of indicators that is impractical. A key statistical question is: how much of the information content (“variation”) of the full indicator set is preserved in a candidate subset?

This analysis applies two methodological approaches to the GII 2025 data (78 Level-1 indicators, 139 countries):

1. **Country Perspective:** Assessing how similar countries are using all indicators versus a subset, using distance matrices and clustering techniques
2. **Variable Perspective:** Using Principal Component Analysis (PCA) to measure how much information a subset captures compared to the full set

2.1 Finding a suitable reduced set

Composite indices such as the GII combine two different measurement logics. At the upper levels - pillars, sub-indices, the index itself - the structure is *formative*: each pillar captures a distinct dimension of innovation performance, and the index’s meaning depends on all of them being represented; dropping a pillar entirely would change what the index measures, not just how precisely it measures it. Within a sub-pillar, by contrast, the indicators are closer to *reflective* measures of a single underlying concept - largely interchangeable manifestations of the same thing, where one or two well-chosen indicators can stand in for the group without changing what is being measured.

This distinction motivates the reduction strategy used throughout this analysis: every pillar and sub-pillar is retained in every candidate subset, preserving the formative structure and therefore the index’s meaning, while indicator reduction happens only *within* each group, by selecting which reflective indicator(s) best represent it. We apply two strategies to decide *how many* reflective indicators to draw from each formative group. The first is to select one indicator per sub-pillar and the second is to select a number per pillar proportional to it’s original size. The metrics for these two strategies ignore correlations between pillars or sub-pillars. However, these cannot be incorporated without also expanding the search to be exhaustive.

We have not implemented an exhaustive search for the best subset. Even restricted to searching *within* a single pillar, exhaustive search grows quickly. Selecting the best possible subset of size k out of p indicators requires evaluating $C(p, k) = \frac{p!}{k!(p-k)!}$ combinations. The Institutions pillar (IN.1) has 6 indicators, so choosing 2 of them already means 15 possible combinations. Multiplied across all 7 pillars, the number of combinations for a subset is prohibitive. This analysis searches exhaustively within each pillar or sub-pillar rather than across the whole index at once. The table below shows how many combinations remain even after allocating a 20-indicator budget proportionally across pillars.

Table 1: Unique combinations per pillar of GII when allocating a 20-indicator budget proportionally

Pillar	Indicators in Pillar	Allocated (budget=20)	Combina- tions
Institutions	6	1	6
Human capital and research	12	3	220
Infrastructure	9	2	36
Market sophistication	11	3	165
Business sophistication	15	4	1365
Knowledge and technology outputs	14	4	1001
Creative outputs	11	3	165

Even after this proportional allocation collapses the problem to one pillar at a time, there are still 1,767,710,124,180,000 combinations to explore in total. Underscoring why within-group exhaustive search is the only tractable approach.

3 Methodology

3.1 Statistical Methods Explanation

The variable perspective uses a subset-selection approach based on Principal Component Analysis (PCA). Rather than simply ranking individual variables by importance, this method evaluates entire subsets of variables according to how well they reproduce the PCA structure of the full dataset.

The criterion, r_m , was introduced by McCabe (1984) as one of several optimality criteria for selecting “principal variables”. In contrast to principle components, principle variables are a subsets of the original variables that best capture variation in the original set. The method was developed in order to select indicators which could function as a principle component while remaining interpretable. Cadima and Jolliffe (2001) formalize this same criterion as one of two indicators for how well a subset of variables approximates a full PCA, and show that r_m^2 is exactly the percentage of total variance in the original data accounted for by regressing the full data on the chosen subset. This is the interpretation of r_m^2 used throughout this report. Both papers show, that choosing variables by their PC loadings alone frequently produces subsets that are suboptimal compared to subsets chosen by directly maximizing r_m .

A key advantage of this method is its built-in protection against redundancy. For example, if multiple indicators are highly correlated, including one in the subset provides most of the information, and adding the others contributes relatively little additional ability to reconstruct the principal components.

The mathematical formula for the selection criterion is:

$$r_m = \sqrt{\frac{\sum_{i=1}^p \lambda_i r_{mi}^2}{\sum_{j=1}^p \lambda_j}}$$

where: - λ_i is the eigenvalue of principal component i (PC_i) from the full dataset, representing the variance explained by that component. The subscript m indexes a candidate subset - r_{mi}^2 is the coefficient of determination (R^2) between principal component i and the candidate subset of variables. Since PC_i is a linear combination of the original variables, the R^2 of the subset measures how well the principal component can be reconstructed from the subset.

Conceptually, this formula computes a weighted average of how well the subset predicts each principal component, with weights given by the variance explained by each component, then normalizes by the total variance. The resulting r_m^2 value can be interpreted as the percentage of total variation explained by the subset, providing an intuitive measure of information retention.

This method assumes that “information” is defined by the variance structure captured by PCA, which is appropriate for statistical data reduction. However, it may downweight variables with

co-variance but high policy relevance, which is an important consideration for substantive index development (c.f.(Algursan et al. 2026)).

3.1.1 The Country Perspective's Distance-Based Criterion

The country perspective asks assess whether a subset preserves how similar countries are judged to be relative to one another. For each candidate subset, the pairwise Euclidean distance between every pair of countries is computed using only that subset's indicators, and the ranking of those pairwise distances is compared to the ranking obtained from the full indicator set, via Spearman rank correlation. A correlation near 1 means the subset reproduces almost the same country-similarity ordering as the full index.

3.2 Proportional Allocation Methodology

Both perspectives select candidate subsets by exhaustive search *within* each pillar or sub-pillar, then assemble the per-group picks into a full subset - consistent with the formative/reflective distinction. Every pillar and sub-pillar is represented in every candidate subset, and only the reflective indicators within each group are subject to selection. This requires deciding how many indicators to take from each group. Two allocation rules are used throughout this report:

- **Flat allocation of 21 indicators:** every pillar contributes (3) indicators, or every sub-pillar contributes 1 indicators. A simple strategy to implement, however, it ignores that GII's pillars differ in the number of indicators, from 6 indicators (Institutions) to 15 (Business sophistication), and its sub-pillars range from 2 to 5 indicators. A flat allocation over-represents small groups and under-represents large ones relative to their actual share of the index. This may limit the subset's utility statistically and leave out conceptual nuances of the GII measurement framework.
- **Proportional allocation of 21 or 42 indicators.** Each group's budget is set proportional to its share of all indicators, using a **largest-remainder (Hamilton) apportionment**. Every group first receives n indicators rounded down to the lowest integer. Any indicators left over are handed out one at a time to the groups with the largest fractional remainder. Unlike flat allocation, a group with a very small share of the index can receive **zero** indicators under this rule. The total budget used is **42** indicators (double the flat allocations' 21). Proportional allocation is applied across pillars and across sub-pillars at this budget, and across pillars again at **21** indicators, matching the flat allocations so that the effect of the allocation rule can be separated from the effect of the budget. No matching sub-pillar case exists: apportioning 21 indicators across 21 sub-pillars returns exactly one indicator each, which is the flat rule. This is also why the larger budget is set above the number of sub-pillars.

Each perspective searches on its own criterion, applied *locally* within a group. The variable perspective maximizes r_m against that group's own sub-PCA rather than the full 78-indicator

PCA. The country perspective maximizes the Spearman rank correlation between that group’s own distance matrix and the candidate subset’s. The allocation rule and the selection criterion are independent choices, so all four allocation rules above are searched under both criteria. This local search is a tractable stand-in for an otherwise infeasible full search. The assembled subset is then re-scored against the full dataset to obtain the metrics reported in this document.

3.3 Data Description

The Global Innovation Index (GII) 2025 dataset ((WIPO) 2024) comprises 78 Level-1 indicators observed for 139 countries/economies, organized in a 5-level hierarchy: 78 indicators -> 21 sub-pillars -> 7 pillars -> 2 sub-indices (Inputs, Outputs) -> 1 overall index. Indicators with negative polarity (e.g. pupil-teacher ratio, applied tariff rate) are reversed before standardization, so that higher standardized values consistently indicate better innovation performance across all indicators. Missing values are imputed via 5-nearest-neighbors using the VIM (Kowarik and Templ 2016) package prior to z-scoring; see the Appendix for imputation statistics.

4 Key Findings

4.1 Country Perspective

4.1.1 Distance Matrix Analysis of Random Subsets

We selected random subset sizes ranging from 10% to 90% of all 78 indicators, while also selecting specifically 42, matching the proportional strategies’ budget below. Each k subset was drawn 1000 times and compared to the full-dataset Euclidean distance matrix via Spearman rank correlation:

Table 2: Rank correlation of random subsets vs. the full distance matrix, by subset size k

Subset Size (k)	Mean Correlation	Median Correlation	Min Correlation	Max Correlation
8	0.678	0.687	0.383	0.824
15	0.784	0.789	0.626	0.878
23	0.847	0.851	0.726	0.918
31	0.891	0.893	0.808	0.936
38	0.920	0.922	0.856	0.955
42	0.932	0.935	0.872	0.962
46	0.946	0.948	0.898	0.971
54	0.963	0.965	0.902	0.983
62	0.978	0.979	0.945	0.991
69	0.988	0.989	0.957	0.996

Subset Size (k)	Mean	Median	Min Correlation	Max
	Correlation	Correlation		Correlation

Correlation rises smoothly and near-monotonically with subset size, from a mean of 0.678 at $k = 8$ to 0.988 at $k = 69$. This random-subset curve is the baseline to the local exhaustive-search strategies.

4.1.2 Flat Allocation

Table 3: Best 3 indicators per pillar, by rank correlation with the full distance matrix

Pillar	Within-Group	
	Rank Correlation	Selected Indicators
Institutions	0.930	Entrepreneurship policies and culture; Operational stability for businesses; Rule of law
Human capital and research	0.775	Gross expenditure on R&D; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	0.734	Electricity output; GDP/unit of energy use; ICT access
Market sophistication	0.728	Domestic credit to private sector; Domestic industry diversification; Joint venture deals
Business sophistication	0.788	Knowledge-intensive employment; Research talent; University-industry R&D collaboration
Knowledge and technology outputs	0.792	High-tech exports; ISO 9001 quality certificates; PCT patents by inventor origin
Creative outputs	0.806	Global brand value, top 5,000; Industrial designs by origin; Top-level domains (TLDs)

Table 4: Best indicator per sub-pillar, by rank correlation with the full distance matrix

Sub-Pillar	Within-Group Rank	
	Correlation	Selected Indicators
Institutional environment	0.900	Operational stability for businesses
Regulatory environment	0.956	Regulatory quality
Business environment	0.771	Policy stability for doing business
Education	0.565	School life expectancy
Tertiary education	0.597	Tertiary enrolment

Sub-Pillar	Within-Group Rank Correlation	Selected Indicators
Research and development (R&D)	0.865	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.839	ICT access
General infrastructure	0.600	Logistics performance
Ecological sustainability	0.528	ISO 14001 environmental certificates
Credit	0.649	Finance for startups and scale-ups
Investment	0.808	Joint venture deals
Trade, diversification and market scale	0.725	Domestic industry diversification
Knowledge workers	0.786	Knowledge-intensive employment
Innovation linkages	0.759	University–industry R&D collaboration
Knowledge absorption	0.664	Research talent
Knowledge creation	0.848	PCT patents by inventor origin
Knowledge impact	0.573	High-tech manufacturing
Knowledge diffusion	0.689	High-tech exports
Intangible assets	0.551	Global brand value, top 5,000
Creative goods and services	0.680	Entertainment and media market
Online creativity	0.942	GitHub commits

Finding: Flat allocation of 3 indicators per pillar reaches a rank correlation of 0.886 with the full distance matrix; flat allocation of 1 indicator per sub-pillar reaches 0.836. The comparable random baseline ($k = 23$) averages 0.847, with a maximum of 0.918 across 1000 draws.

The two allocation levels therefore diverge. Pillar-level allocation exceeds the random average by 0.039, while sub-pillar-level allocation falls 0.011 below it. Neither reaches the best random subset of the same size, which is expected: the search is confined within groups, so it cannot exploit the cross-group combinations available to an unconstrained draw. The constraint is accepted deliberately, since it is what guarantees every pillar and sub-pillar remains represented and the index retains its formative meaning.

4.1.3 Clustering Results: SIMPROF + Rand Index

Against the full-data country clustering, flat allocation by pillar recovers a Rand Index of 0.891 and flat allocation by sub-pillar 0.887. Both subsets therefore reproduce the full clustering to within roughly 11% of country pairs, and the two allocation levels are indistinguishable on this measure despite their 0.05 gap in rank correlation.

Table 5: Rand Index: full-data clustering vs. flat-allocation clustering (by pillar and by sub-pillar)

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Flat allocation (pillar)	0.891	78	21	Flat allocation (pillar)
Full vs. Flat allocation (sub-pillar)	0.887	78	21	Flat allocation (sub-pillar)

4.1.4 Proportional Allocation

We proportionally allocate 42 indicators, using the largest-remainder method described in Methodology above.

Table 6: Proportional allocation across pillars

Pillar	Indicators Allocated	Within-Group Rank Correlation	Selected Indicators
Institutions	3	0.930	Entrepreneurship policies and culture; Operational stability for businesses; Rule of law
Human capital and research	6	0.889	Government funding/pupil, secondary; Pupil-teacher ratio, secondary; QS university ranking; Gross expenditure on R&D; School life expectancy; Tertiary inbound mobility
Infrastructure	5	0.873	GDP/unit of energy use; Government’s online service; ICT use; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.935	Applied tariff rate, weighted avg.; Domestic credit to private sector; Finance for startups and scale-ups; Domestic industry diversification; Joint venture deals; Loans from microfinance institutions

Pillar	Indicators Allocated	Within- Group Rank Correlation	Selected Indicators
Business sophistication	8	0.946	GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Knowledge-intensive employment; Patent families; Research talent; State of cluster development
Knowledge and technology outputs	8	0.949	Software spending; Labor productivity growth; High-tech exports; Intellectual property receipts; ISO 9001 quality certificates; PCT patents by inventor origin; Scientific and technical articles; Unicorn valuation
Creative outputs	6	0.923	Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; GitHub commits; Industrial designs by origin; Trademarks by origin

Table 7: Proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	Within- Group Rank Correlation	Selected Indicators
Institutional environment	1	0.900	Operational stability for businesses
Regulatory environment	1	0.956	Regulatory quality
Business environment	1	0.771	Policy stability for doing business
Education	3	0.866	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.855	Graduates in science and engineering; Tertiary enrolment
Research and development (R&D)	2	0.968	QS university ranking; Researchers
Information and communication technologies (ICTs)	2	0.962	Government's online service; ICT access

Sub-Pillar	Indicators Allocated	Within- Group Rank Correlation	Selected Indicators
General infrastructure	2	0.864	Gross capital formation; Logistics performance
Ecological sustainability	1	0.528	ISO 14001 environmental certificates
Credit	1	0.649	Finance for startups and scale-ups
Investment	3	0.983	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.725	Domestic industry diversification
Knowledge workers	3	0.940	Emerging knowledge workforce; Females employed w/advanced degrees; GERD performed by business
Innovation linkages	3	0.930	Public research–industry co-publications; State of cluster development; WU Rankings industry and global outlook
Knowledge absorption	3	0.917	High-tech imports; Intellectual property payments; Research talent
Knowledge creation	3	0.989	Citable documents H-index; PCT patents by inventor origin; Scientific and technical articles
Knowledge impact	2	0.760	Labor productivity growth; High-tech manufacturing
Knowledge diffusion	3	0.909	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.833	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.843	Creative goods exports; Entertainment and media market
Online creativity	1	0.942	GitHub commits

Finding: At the 42-indicator budget, proportional allocation across pillars achieves a rank correlation of 0.955, against 0.907 across sub-pillars. The random baseline at this size averages 0.932.

The advantage of pillar-level over sub-pillar-level allocation seen at 21 indicators persists at 42, and in the same direction: pillar-level allocation again exceeds the random average while

sub-pillar-level allocation falls below it. Doubling the budget raises every figure, but does not alter the ordering between the two allocation levels.

Applying the same proportional rule across pillars at the flat 21-indicator budget returns 0.887, against 0.886 for flat allocation at that budget. Separating the two effects this way shows the gain to be a function of the budget rather than of the rule.

Table 8: Rand Index: full clustering vs. proportional-allocation subset clustering

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Proportional allocation (pillar)	0.89	78	42	Proportional allocation (pillar)

Finding: The 42-indicator proportional subset recovers a Rand Index of 0.89, against 0.891 and 0.887 for the two 21-indicator flat subsets.

All three values fall within 0.004 of one another, even though the subsets differ in size by a factor of two and span 0.119 in rank correlation. Recovery of the country clustering is therefore insensitive to both subset size and rank correlation over this range: once every pillar and sub-pillar is represented, additional indicators sharpen the distance structure without redrawing the cluster boundaries.

4.2 Variable Perspective

4.2.1 PCA Analysis

The table below shows the number of principal components needed to capture specified levels of variance in the full 78-indicator dataset:

Table 9: Components needed for target variance levels (full PCA)

Variance Level	Components Needed
80%	17
90%	26
95%	36

Finding: The first 17 of 78 components already capture 80% of total variance, confirming substantial redundancy in the full indicator set - a necessary condition for indicator reduction to be viable at all. Note this is a property of the full PCA (unconstrained linear combinations of all 78 indicators), not directly comparable to how many *raw indicators* a subset needs.

4.2.2 Reflective Indicators per Sub-Pillar (Flat Allocation - Headline Subset)

In this section we present results for selecting within each sub-pillar the indicator with the best local r_m .

Table 10: Flat allocation (sub-pillar): reflective indicator per sub-pillar

Sub-Pillar	r_m of Best Subset	Selected Indicators
Institutional environment	0.938	Operational stability for businesses
Regulatory environment	0.975	Rule of law
Business environment	0.834	Entrepreneurship policies and culture
Education	0.626	School life expectancy
Tertiary education	0.615	Tertiary enrolment
Research and development (R&D)	0.793	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.877	ICT access
General infrastructure	0.647	Logistics performance
Ecological sustainability	0.588	ISO 14001 environmental certificates
Credit	0.650	Finance for startups and scale-ups
Investment	0.750	Joint venture deals
Trade, diversification and market scale	0.639	Domestic industry diversification
Knowledge workers	0.802	Knowledge-intensive employment
Innovation linkages	0.727	University–industry R&D collaboration
Knowledge absorption	0.508	Research talent
Knowledge creation	0.638	PCT patents by inventor origin
Knowledge impact	0.592	High-tech manufacturing
Knowledge diffusion	0.649	Production and export complexity
Intangible assets	0.613	Industrial designs by origin
Creative goods and services	0.550	Entertainment and media market
Online creativity	0.639	GitHub commits

Finding: This 21-indicator subset achieves $r_m = 0.826$, i.e. r_m^2 68.2% of the full dataset’s

PCA variance structure.

4.2.3 Flat Allocation by Pillar (3 per Pillar) - Comparison

Table 11: Flat allocation (pillar): best 3 indicators per pillar

Pillar	r_m of Best Subset	Selected Indicators
Institutions	0.965	Entrepreneurship policies and culture; Policy stability for doing business; Regulatory quality
Human capital and research	0.757	Expenditure on education; Gross expenditure on R&D; Tertiary enrolment
Infrastructure	0.784	Gross capital formation; ICT access; Low-carbon energy use
Market sophistication	0.764	Domestic credit to private sector; Domestic market scale; Joint venture deals
Business sophistication	0.746	Females employed w/advanced degrees; High-tech imports; University–industry R&D collaboration
Knowledge and technology outputs	0.702	Software spending; Production and export complexity; Patents by origin
Creative outputs	0.697	Entertainment and media market; National feature films; Industrial designs by origin

Finding: Flat allocation by pillar reaches $r_m = 0.854$ ($r_m^2 = 72.9\%$) - marginally *higher* than flat allocation by sub-pillar’s 0.826 at the same 21-indicator size.

4.2.4 Proportional Allocation by Pillar and Sub-Pillar - Comparison

Table 12: Proportional allocation across pillars

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutions	3	0.965	Entrepreneurship policies and culture; Policy stability for doing business; Regulatory quality

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Human capital and research	6	0.898	Expenditure on education; Global corporate R&D investors; Researchers; Graduates in science and engineering; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	5	0.900	GDP/unit of energy use; Gross capital formation; ICT access; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.926	Applied tariff rate, weighted avg.; Domestic industry diversification; Joint venture deals; Late VC deals intensity; Market capitalization; Loans from microfinance institutions
Business sophistication	8	0.932	Emerging knowledge workforce; FDI net inflows; GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Patent families; State of cluster development
Knowledge and technology outputs	8	0.915	Software spending; Labor productivity growth; High-tech exports; ICT services exports; ISO 9001 quality certificates; PCT patents by inventor origin; Unicorn valuation; Utility models by origin
Creative outputs	6	0.865	Mobile app creation; Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; National feature films; Trademarks by origin

Table 13: Proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutional environment	1	0.938	Operational stability for businesses
Regulatory environment	1	0.975	Rule of law
Business environment	1	0.834	Entrepreneurship policies and culture
Education	3	0.882	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.840	Graduates in science and engineering; Tertiary inbound mobility

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Research and development (R&D)	2	0.922	Global corporate R&D investors; Researchers
Information and communication technologies (ICTs)	2	0.966	Government's online service; ICT use
General infrastructure	2	0.867	Electricity output; Gross capital formation
Ecological sustainability	1	0.588	ISO 14001 environmental certificates
Credit	1	0.650	Finance for startups and scale-ups
Investment	3	0.980	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.639	Domestic industry diversification
Knowledge workers	3	0.944	Females employed w/advanced degrees; GERD financed by business; GERD performed by business
Innovation linkages	3	0.937	Patent families; Public research-industry co-publications; State of cluster development
Knowledge absorption	3	0.813	High-tech imports; ICT services imports; Intellectual property payments
Knowledge creation	3	0.907	PCT patents by inventor origin; Scientific and technical articles; Utility models by origin
Knowledge impact	2	0.777	Labor productivity growth; High-tech manufacturing
Knowledge diffusion	3	0.871	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.811	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.739	Creative goods exports; National feature films
Online creativity	1	0.639	GitHub commits

Finding: At 42 indicators, proportional allocation by pillar reaches $r_m = 0.963$ ($r_m^2 = 92.6\%$) and proportional allocation by sub-pillar reaches $r_m = 0.946$ ($r_m^2 = 89.5\%$) - both a large step up from the 21-indicator flat allocations, as expected with double the budget.

4.2.5 Flat-Allocation Sweep: 1-4 Indicators per Pillar

To see how r_m grows as the flat per-pillar allocation increases, the flat allocation by pillar search was repeated for 1, 2, 3, and 4 indicators per pillar:

Table 14: r_m as the flat per-pillar allocation increases

Indicators per Pillar	r_m	Total Indicators
1	0.705	7
2	0.802	14
3	0.854	21
4	0.900	28

Finding: r_m rises from 0.705 at 7 variables (1/pillar) to 0.9 at 28 variables (4/pillar) - a steep gain per additional variable early on, consistent with the PCA finding above that a small number of dimensions already captures most of the variance structure.

4.2.6 Random-Subset r_m Curve vs. #PCs Needed

For the same subset sizes used in the country perspective's Fig 2, r_m was computed for 20 random subsets per size. r_m is far more expensive per evaluation than the rank-correlation metric. It requires regressing all 78 principal components on the candidate subset, so fewer random draws were used per size here than Table 2's 1,000.

Table 15: Random-subset r_m by size, vs. cumulative variance explained by the first k principal components

Subset Size (k)	r_m	Subset Variance Captured	PCA Variance (k components)
8	0.695	0.484	0.649
15	0.791	0.626	0.785
23	0.857	0.734	0.878
31	0.900	0.811	0.931
38	0.925	0.856	0.959
42	0.941	0.885	0.969
46	0.951	0.904	0.978
54	0.969	0.940	0.989
62	0.983	0.966	0.996
69	0.992	0.984	0.999

Finding: At $k = 21$ (interpolating between $k = 15$ and $k = 23$ above), a random subset averages roughly the same r_m as the exhaustive flat-allocation results report (0.854 by pillar /

0.826 by sub-pillar) - so at this budget the constrained search offers no advantage over chance on the PCA criterion, in contrast to the country perspective, where pillar-level allocation exceeded its random baseline. The PCA-variance column is consistently higher than a same-sized *indicator* subset's r_m^2 at every k. Since the first k principal components are unconstrained linear combinations of all 78 indicators, a subset is restricted to k raw indicators. The resulting gap in explained variance is the cost of requiring the final subset to be interpretable indicators rather than statistical composites.

4.2.7 Headline Recommended Subset

The 21-indicator subset from flat allocation by sub-pillar above, with full indicator names:

Table 16: Headline subset: one reflective indicator per sub-pillar

Sub-Pillar	Indicator	r_m of Best Subset
Institutional environment	Operational stability for businesses	0.938
Regulatory environment	Rule of law	0.975
Business environment	Entrepreneurship policies and culture	0.834
Education	School life expectancy	0.626
Tertiary education	Tertiary enrolment	0.615
Research and development (R&D)	Gross expenditure on R&D	0.793
Information and communication technologies (ICTs)	ICT access	0.877
General infrastructure	Logistics performance	0.647
Ecological sustainability	ISO 14001 environmental certificates	0.588
Credit	Finance for startups and scale-ups	0.650
Investment	Joint venture deals	0.750
Trade, diversification and market scale	Domestic industry diversification	0.639
Knowledge workers	Knowledge-intensive employment	0.802
Innovation linkages	University–industry R&D collaboration	0.727
Knowledge absorption	Research talent	0.508
Knowledge creation	PCT patents by inventor origin	0.638
Knowledge impact	High-tech manufacturing	0.592

Sub-Pillar	Indicator	r_m of Best Subset
Knowledge diffusion	Production and export complexity	0.649
Intangible assets	Industrial designs by origin	0.613
Creative goods and services	Entertainment and media market	0.550
Online creativity	GitHub commits	0.639

Finding: This subset captures r_m^2 68.2% of the full dataset’s PCA variance structure and recovers a Rand Index of 0.887 against the full-data country clustering, while cutting the indicator count from 78 to 21 (a 73% reduction).

5 Cross-Allocation Comparison

Table 17: Country perspective: allocation rules compared, by budget

Allocation Rule	Indicators	Rank Correlation
Flat allocation (pillar)	21	0.886
Flat allocation (sub-pillar)	21	0.836
Proportional allocation (pillar)	21	0.887
Proportional allocation (pillar)	42	0.955
Proportional allocation (sub-pillar)	42	0.907

Table 18: Variable perspective: allocation rules compared, by budget

Allocation Rule	Indicators	r_m	r_m^2 (%)
Flat allocation (pillar)	21	0.854	72.9
Flat allocation (sub-pillar, headline)	21	0.826	68.2
Proportional allocation (pillar)	21	0.860	73.9
Proportional allocation (pillar)	42	0.963	92.6
Proportional allocation (sub-pillar)	42	0.946	89.5

Finding: Proportional allocation at 42 indicators outperforms flat allocation at 21 indicators on both metrics. Compared at a common budget, however, the two rules are nearly indistinguishable: at 21 indicators across pillars, proportional allocation reaches a rank correlation of 0.887 against flat allocation’s 0.886, and $r_m = 0.86$ against 0.854.

Budget therefore accounts for almost all of the apparent advantage. Holding the rule fixed and doubling the budget from 21 to 42 indicators raises the rank correlation by 0.068 and r_m by 0.103; holding the budget fixed and switching from flat to proportional changes them by 0.001 and 0.006. The budget effect is roughly two orders of magnitude larger on the distance metric and one on the PCA metric.

The case for proportional allocation is therefore representational rather than statistical: no group is systematically over- or under-weighted relative to its size in the index, at no measurable cost to either criterion. Whether indicator count is an appropriate basis for that weighting remains a judgement for the analyst, and the near-equivalence observed here need not hold for an index whose sub-pillars are more strongly correlated within pillars.

6 Conclusions

1. **Meaningful reduction is possible, with a clear methodological trade-off.** The headline 21-indicator subset (one reflective indicator per sub-pillar) captures r_m^2 68.2% of the full PCA’s variance structure and a Rand Index of 0.887 against the full clustering, at a 73% reduction in indicator count.
2. **The gain over random selection depends on the level at which indicators are allocated, not on the budget.** Allocating at pillar level exceeds the average random subset of matched size at both budgets (0.886 against 0.847 at 21 indicators; 0.955 against 0.932 at 42). Allocating at sub-pillar level falls below it at both. No constrained subset reaches the best random subset of its size, which is the cost of requiring every group to be represented.
3. **The allocation rule matters far less than the budget.** Compared at a common 21-indicator budget, flat and proportional allocation across pillars differ by 0.001 in rank correlation and 0.006 in r_m , while doubling the budget under a fixed rule gains 0.068 and 0.103 respectively. Proportional allocation’s benefit is representational rather than statistical.
4. **Recovery of the country clustering saturates well below the full indicator set.** The three candidate subsets return Rand Indices within 0.004 of one another despite differing twofold in size and by 0.119 in rank correlation. Beyond covering every pillar and sub-pillar, further indicators refine the distance structure without changing which countries group together.
5. **The r_m criterion treats “information” as PCA variance, which may downweight indicators with low covariance but high substantive/policy relevance; the headline subset should be reviewed against domain expertise, not adopted on statistical grounds alone.**

7 Limitations

1. **Single-year analysis.** This analysis covers GII 2025 only; temporal stability of the recommended subset across years has not been assessed.
2. **PCA-defined “information.”** The r_m criterion equates information content with PCA variance structure; a substantively important but low covariance indicator could be excluded even if policy-relevant.
3. **Local search proxy.** Exhaustive search within each pillar/sub-pillar uses a local r_m computed on that group’s own sub-PCA (not the full 78-indicator PCA) as a tractable stand-in for an otherwise infeasible full search; the assembled subset is re-scored against the true full-dataset metric for reporting, but the *search* itself never directly optimizes that metric.
4. **Linear relationships assumed.** Both the rank-correlation and r_m criteria are built on linear/monotonic structure (PCA, Euclidean distance); non-linear relationships between indicators are not captured.
5. **Domain-specific validity not tested.** Whether the recommended subset remains meaningful to GII’s intended policy audience has not been evaluated here and would require domain-expert review.

8 Data Availability

All data and code are available in this repository:

- Raw data: `gii-data/idata.xlsx`, `gii-data/imeta.xlsx`
- Processed data: `data/gii_data_standardized.csv`, `data/gii_data_complete_cases.csv`
- Analysis scripts: `scripts/00_setup.R` through `scripts/04_variable_perspective.R`, shared functions in `scripts/R/`
- Results: `outputs/` (CSVs for tabular results, `.rds` for clustering objects)

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10 Appendix

11 Data Imputation Documentation

Missing data is an inherent challenge in composite index construction. This analysis used k-Nearest Neighbors (kNN) imputation to address missing values before conducting the main analyses.

11.1 Imputation Process

- **Method:** k-Nearest Neighbors (kNN) imputation
- **Parameters:** k = 5 neighbors (VIM: :kNN, other parameters left at package defaults)
- **Implementation:** R VIM package
- **Indicators Used:** Level 1 indicators from IMETA metadata

11.2 Missing Data Statistics

Table 19: Imputation Process Summary

Metric	Count
Total Missing Values	1211
Variables with Missing Data	58
Countries with Missing Data	135

Note: 1211 missing values across 58 of 78 indicators, affecting 135 of 139 countries, were imputed prior to standardization.

12 Descriptive Statistics Tables

12.1 Descriptive Statistics - Original Data

The table below presents descriptive statistics for all Level 1 indicators in the original dataset before imputation:

Table 20: Descriptive Statistics - Original Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Government effectiveness	0.15	0.03	-1.75	2.32	-0.71	0.29
Operational stability for businesses	2.23	2.15	0.70	4.45	0.03	0.58
Regulatory quality	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law	0.07	-0.12	-2.15	1.97	-0.77	0.29
Entrepreneurship policies and culture	4.72	4.72	2.92	7.25	-0.21	0.45
Policy stability for doing business	4.03	3.98	1.83	6.45	-0.69	0.02
Expenditure on education	4.25	4.16	0.35	9.39	0.80	0.29
Government funding/pupil, secondary	19.60	19.41	6.00	51.06	1.57	0.74
PISA scales in reading, maths and science	439.44	439.29	337.45	579.03	-0.84	0.07
Pupil-teacher ratio, secondary	15.89	13.53	6.14	68.13	9.84	2.46
School life expectancy	14.17	14.36	6.34	20.85	-0.23	-0.28
Graduates in science and engineering	23.29	23.14	9.56	41.10	-0.33	0.17
Tertiary enrolment	51.51	54.40	2.71	166.67	-0.02	0.37

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Tertiary inbound mobility	7.34	4.24	0.00	69.73	15.40	3.35
QS university ranking	21.29	10.43	0.00	97.63	0.29	1.14
Global corporate R&D investors	926.35	0.00	0.00	36142.79	60.93	7.11
Gross expenditure on R&D	1.01	0.61	0.01	6.35	4.50	1.92
Researchers	2269.70	922.47	4.50	10413.00	0.53	1.21
Government's online service	0.68	0.72	0.17	1.00	-0.95	-0.40
ICT access	83.02	88.25	24.33	100.00	0.54	-1.12
ICT use	79.65	84.47	29.84	99.36	1.78	-1.49
Electricity output	4936.33	3183.45	29.76	53263.16	24.28	4.06
Gross capital formation	24.49	23.70	11.60	44.56	0.63	0.74
Logistics performance	3.12	3.10	2.10	4.30	-1.10	0.19
GDP/unit of energy use	12.23	11.06	2.16	41.21	4.01	1.59
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
Domestic credit to private sector	60.33	48.53	8.14	248.76	2.51	1.51
Finance for startups and scale-ups	4.49	4.49	1.99	6.90	-0.48	0.09
Loans from microfinance institutions	2.09	1.05	0.00	17.34	12.83	3.33
Joint venture deals	0.21	0.06	0.00	4.55	56.21	6.66
Late VC deals intensity	0.32	0.03	0.00	11.43	55.80	7.13
Market capitalization	80.85	37.45	0.20	1507.77	55.74	6.96
Venture capital investors	0.65	0.14	0.01	29.23	101.26	9.79
Venture capital recipients	0.24	0.08	0.01	5.50	67.12	7.42
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Domestic industry diversification	0.18	0.14	0.07	0.55	3.07	1.61
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Females employed w/advanced degrees	13.49	12.78	0.14	31.04	-1.14	0.26
GERD financed by business	34.31	38.44	0.00	80.84	-1.07	-0.07
GERD performed by business	0.77	0.38	0.00	5.90	7.67	2.33
Knowledge-intensive employment	27.07	23.94	2.02	68.01	-0.95	0.31
Patent families	1.30	0.03	0.00	73.67	114.09	10.30
Public research-industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
State of cluster development	4.00	3.90	2.59	5.40	-0.79	0.24
WU Rankings industry and global outlook	55.74	49.40	26.70	97.10	-0.99	0.58

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
University–industry R&D collaboration	3.97	3.86	2.30	6.72	-0.12	0.38
FDI net inflows	3.93	2.50	- 161.34	224.25	66.45	3.27
High-tech imports	9.33	8.01	1.30	55.19	21.43	3.65
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
Research talent	33.25	32.27	0.30	81.91	-1.19	0.21
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Patents by origin	2.54	0.62	0.00	61.56	39.40	5.92
PCT patents by inventor origin	0.62	0.07	0.00	7.57	11.91	3.30
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Utility models by origin	1.55	0.16	0.00	88.51	84.62	9.17
Software spending	0.24	0.21	0.00	1.26	3.08	1.42
Labor productivity growth	0.83	0.86	-4.57	6.26	0.84	-0.01
High-tech manufacturing	25.19	21.99	1.04	79.65	0.15	0.75
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Production and export complexity	0.09	-0.05	-2.18	2.29	-0.52	0.23
High-tech exports	4.57	1.51	0.02	48.39	12.41	3.08
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Global brand value, top 5,000	3.59	0.78	0.00	21.12	1.78	1.64
Intangible asset intensity, top 15	41.66	56.72	- 371.30	89.79	28.96	-4.81
Industrial designs by origin	2.21	0.92	0.00	23.28	11.87	3.25
Trademarks by origin	36.72	28.95	0.00	202.33	7.36	2.24
Creative goods exports	1.27	0.38	0.00	11.50	7.00	2.60
Cultural and creative services exports	0.96	0.46	0.00	18.31	53.87	6.48
Entertainment and media market	0.93	0.47	0.02	3.59	-0.06	0.92
National feature films	5.01	2.67	0.00	46.93	18.95	3.70
Mobile app creation	1379627.73	181351.44	0.04	98744342.48	123.62	10.98
GitHub commits	1326116.27	410927.87	2571.55	14402260.97	14.47	2.99
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93

12.2 Descriptive Statistics - Complete Data

The table below presents descriptive statistics for all Level 1 indicators in the complete dataset after kNN imputation and case-wise cleaning:

Table 21: Descriptive Statistics - Complete Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Operational stability for businesses	2.23	2.15	0.70	4.45	0.03	0.58
Government effectiveness	0.15	0.03	-1.75	2.32	-0.71	0.29
Regulatory quality	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law	0.07	-0.12	-2.15	1.97	-0.77	0.29
Policy stability for doing business	4.03	3.98	1.83	6.45	-0.65	0.02
Entrepreneurship policies and culture	4.63	4.54	2.92	7.25	0.28	0.64
Expenditure on education	4.24	4.14	0.35	9.39	0.83	0.31
Government funding/pupil, secondary	18.48	17.96	6.00	51.06	1.96	0.85
School life expectancy	13.92	13.94	6.34	20.85	-0.42	-0.13
PISA scales in reading, maths and science	419.11	403.60	337.45	579.03	-0.43	0.70
Pupil-teacher ratio, secondary	15.80	13.28	6.14	68.13	10.01	2.48
Tertiary enrolment	50.58	53.42	2.71	166.67	0.00	0.41
Graduates in science and engineering	22.88	22.88	9.56	41.10	-0.31	0.25
Tertiary inbound mobility	6.61	3.14	0.00	69.73	17.76	3.58
Researchers	1922.18	669.77	4.50	10413.00	1.08	1.44
Gross expenditure on R&D	0.88	0.42	0.01	6.35	5.71	2.16
Global corporate R&D investors	933.55	0.00	0.00	36142.79	61.25	7.12
QS university ranking	21.29	10.43	0.00	97.63	0.29	1.14
ICT access	82.88	88.19	24.33	100.00	0.47	-1.09
ICT use	78.81	83.81	29.84	99.36	1.28	-1.36
Government's online service	0.68	0.73	0.17	1.00	-0.94	-0.41
Electricity output	4663.61	2879.19	29.76	53263.16	25.59	4.17
Logistics performance	3.04	2.90	2.10	4.30	-0.96	0.45
Gross capital formation	24.48	23.70	11.60	44.56	0.67	0.75
GDP/unit of energy use	12.00	10.93	2.16	41.21	4.46	1.69
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Finance for startups and scale-ups	4.25	4.11	1.99	6.90	-0.45	0.48
Domestic credit to private sector	60.08	48.30	8.14	248.76	2.47	1.51
Loans from microfinance institutions	1.43	0.63	0.00	17.34	26.25	4.59
Market capitalization	58.57	25.34	0.20	1507.77	88.80	8.73
Venture capital investors	0.56	0.11	0.01	29.23	120.91	10.68
Venture capital recipients	0.22	0.07	0.01	5.50	73.92	7.78
Late VC deals intensity	0.25	0.01	0.00	11.43	71.99	8.09
Joint venture deals	0.18	0.04	0.00	4.55	66.13	7.21
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic industry diversification	0.19	0.17	0.07	0.55	2.05	1.19
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Knowledge-intensive employment	26.67	23.31	2.02	68.01	-0.89	0.38
Females employed w/advanced degrees	13.06	11.20	0.14	31.04	-1.04	0.35
GERD performed by business	0.50	0.06	0.00	5.90	11.94	2.97
GERD financed by business	28.45	28.73	0.00	80.84	-1.02	0.34
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
University–industry R&D collaboration	3.95	3.83	2.30	6.72	-0.05	0.42
WU Rankings industry and global outlook	51.65	42.20	26.70	97.10	-0.25	1.02
State of cluster development	3.99	3.84	2.59	5.40	-0.75	0.28
Patent families	1.29	0.03	0.00	73.67	114.91	10.33
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
High-tech imports	9.32	8.03	1.30	55.19	21.61	3.67
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
FDI net inflows	3.91	2.49	- 161.34	224.25	66.93	3.29
Research talent	23.49	11.34	0.30	81.91	-0.61	0.86
Patents by origin	2.51	0.61	0.00	61.56	39.97	5.96
PCT patents by inventor origin	0.58	0.06	0.00	7.57	13.22	3.45
Utility models by origin	1.05	0.16	0.00	88.51	136.59	11.64
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Labor productivity growth	0.83	0.86	-4.57	6.26	0.87	0.00
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Software spending	0.24	0.20	0.00	1.26	3.09	1.42
High-tech manufacturing	22.20	18.35	1.04	79.65	0.37	0.94
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
Production and export complexity	0.08	-0.06	-2.18	2.29	-0.43	0.26
High-tech exports	4.53	1.51	0.02	48.39	12.49	3.09
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Intangible asset intensity, top 15	32.37	39.35	-	89.79	34.44	-4.68
			371.30			
Trademarks by origin	36.74	28.99	0.00	202.33	7.57	2.26
Global brand value, top 5,000	3.20	0.47	0.00	21.12	2.47	1.81
Industrial designs by origin	2.12	0.84	0.00	23.28	12.62	3.34
Cultural and creative services exports	0.86	0.42	0.00	18.31	60.72	6.87
National feature films	4.47	2.69	0.00	46.93	26.15	4.26
Entertainment and media market	0.54	0.17	0.02	3.59	2.74	1.84
Creative goods exports	1.27	0.38	0.00	11.50	7.07	2.61
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93
GitHub commits	1357449.63	411718.75	2571.55	14402260.97	13.92	2.91
Mobile app creation	1305696.24	155386.88	0.04	98744342.48	131.12	11.31